

Cyclistic: Analysis on Rider Behaviors

Executive Summary

Cyclistic's strategic focus has shifted from market expansion to increasing profitability, with annual members identified as significantly more profitable than casual riders. This analysis examines behavioral differences between these two groups to inform marketing strategies for conversion of casual users into members.

Analysis of 2025 rider data shows that while casual riders take fewer trips overall, they consistently record longer ride durations, particularly on weekends and during summer months, averaging 42% shorter rides throughout the year. In contrast, members exhibit more frequent and consistent usage throughout the year, averaging roughly 1.5 million more total rides. This suggests that casual riders use the service more sporadically and may perceive higher per-ride costs, while members integrate the service into regular routines.

Based on these insights, the most effective conversion strategy is to emphasize the financial value of membership. Highlighting cost savings over time and the benefit of unlimited access can remove barriers for financial friction for conversion, and position membership as the more economical and convenient option for casual riders. Utilizing these marketing campaigns during the high usership periods of Q2 and Q3 would likely yield the highest potential conversion and market base increase.

Business Problem

The central question guiding this analysis: “How do annual members and casual riders use Cyclistic bikes differently?”

Understanding how casual and membered riders utilize the service Cyclistic offers is essential for identifying targeted marketing strategies to convert riders into annual members. This analysis aims to describe usage behaviors among rider populations to inform the marketing team in how best to structure and implement more effective marketing campaigns directed towards increasing this conversion.

Data Sources

Rider data for each month of 2025 was utilized in this analysis. The data size for each month ranges from approximately 100,000 to 800,000 entries, depending on seasonal fluctuations.

Each data set contains key variables such as ride ID, rideable type, start and end timestamps, station locations, and rider classification (member or casual).

The primary limitation is the lack of unique rider identifiers. There is no information that allows the linkage of rides to individual users, preventing the analysis of rider level behaviors, such as individual ride patterns and usage variability. This limitation forces this analysis to be limited to aggregate comparisons between casual and membership riders.

Additionally, the data set does not account for holidays, weather fluctuations, or local events, which can have an influence in short term fluctuation in rider behavioral patterns.

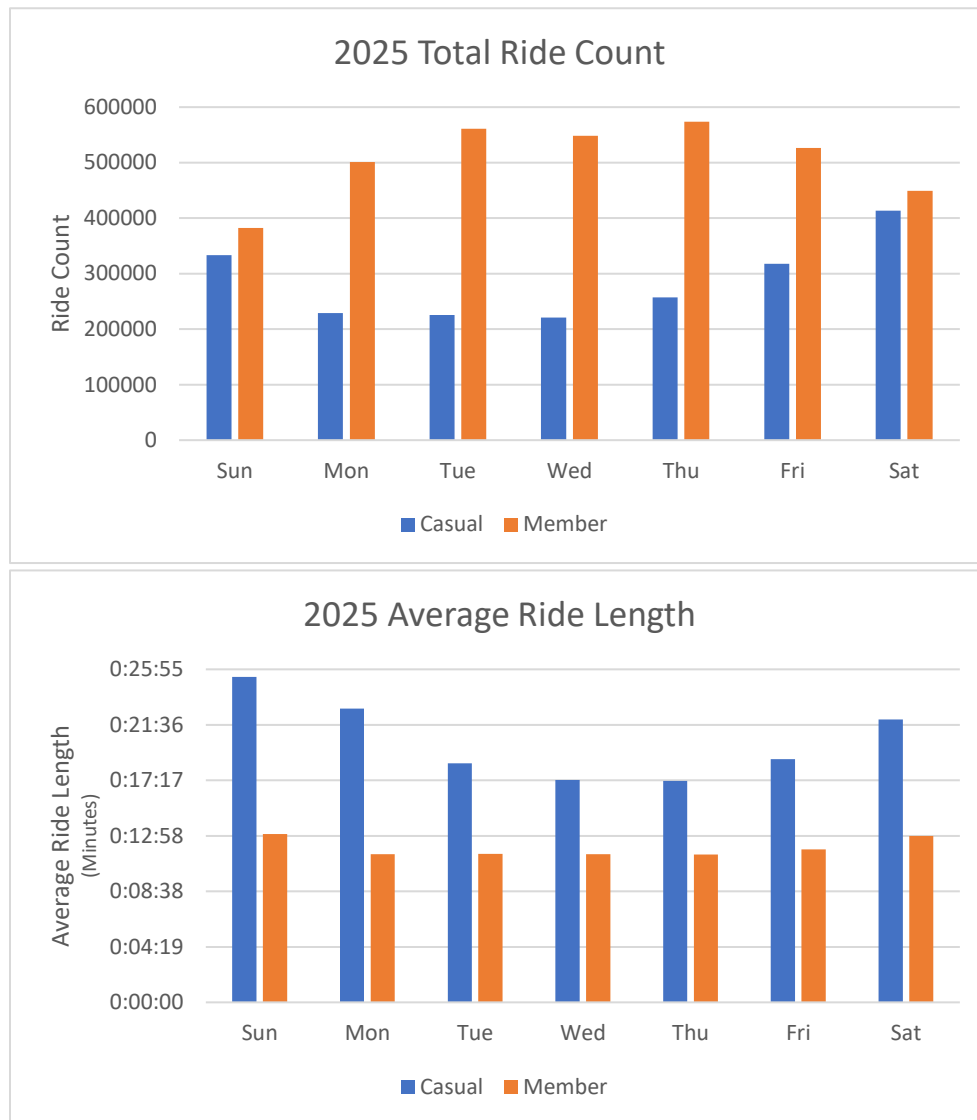
Data Preparation and Cleaning

Data preparation focused on ensuring consistency and useability across each monthly data set. Fields were standardized to conform to the appropriate data type to ensure accurate analysis.

Two fields were created within each data set to allow for deeper insights: Ride duration (Calculated as the difference between end and start times) and day of the week (Generated using Excel’s weekday function) to allow for further analysis of usage patterns across the week.

Basic data validation was performed to ensure that all records utilized in the analysis were valid and contained the required information, to ensure consistency and reliability across metrics.

Total Yearly Analysis



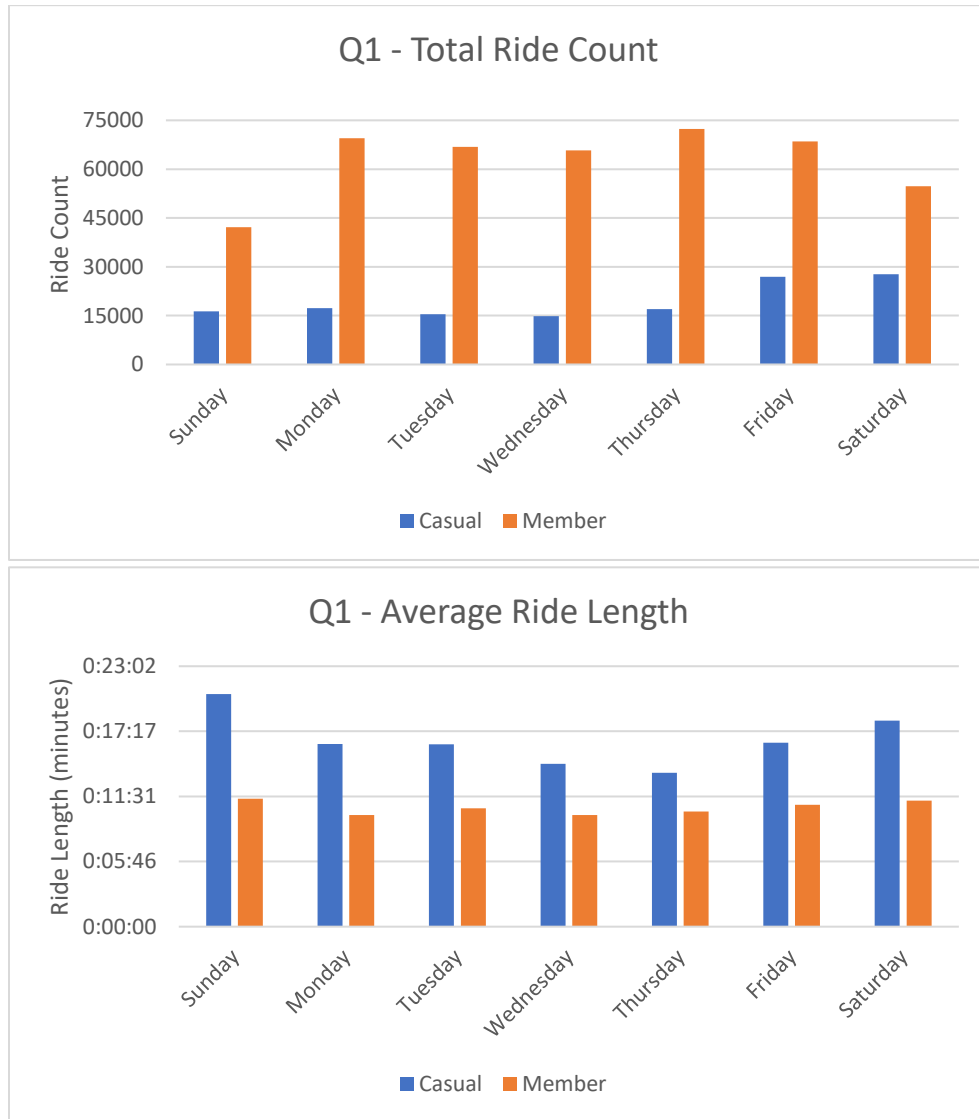
Clear behavioral differences emerged between ridership populations among the two key variables that were looked at: ride frequency and average ride duration.

Members take significantly more rides overall, roughly 44% more total rides than casual riders, while maintaining shorter and more consistent ride durations across all days of the week. Casual riders take fewer and less routine rides, but maintain notably longer ride durations, about 69% longer average ride times, with increasing usage occurring on weekends.

These patterns suggest that members primarily use the service for routine, utility-based travel, while casual riders are more likely to use it for infrequent and recreational reasons. The longer ride durations among casual riders may reflect a desire for value per ride, where members appear less constraint by per-ride cost.

Quarterly Analysis

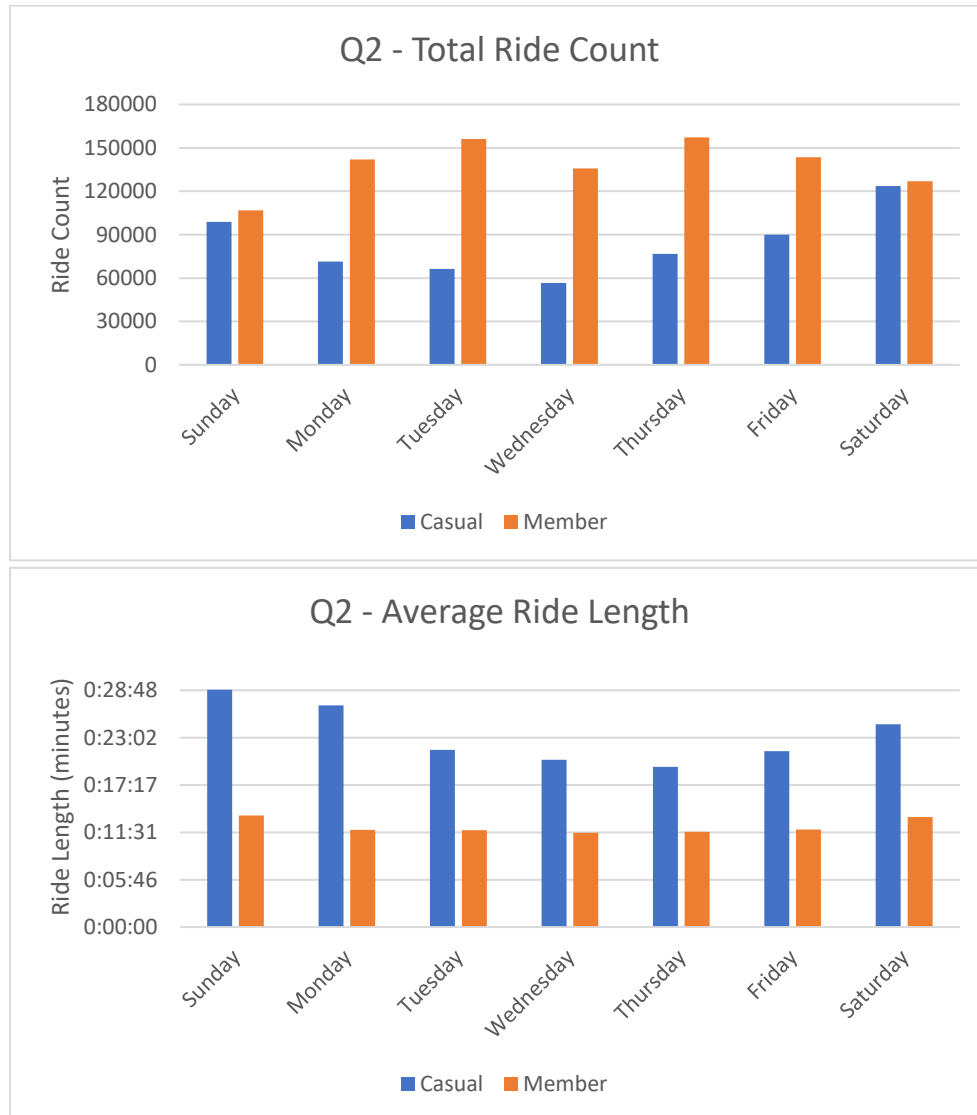
Q1



Ridership is lowest in Q1, driven primarily by a sharp decline in casual usage, while member ride frequency remains relatively stable. Member rides in Q1 make up roughly 12.4% of total yearly rides, and casual rides only about 7% of total yearly rides. Members continue to exhibit consistent weekday usage, whereas casual riders maintain a heavy weekend pattern at reduced volumes.

This suggests that member usage is less sensitive to seasonal conditions, although does still decrease in the same time periods as casual usage, albeit at a less dramatic rate.

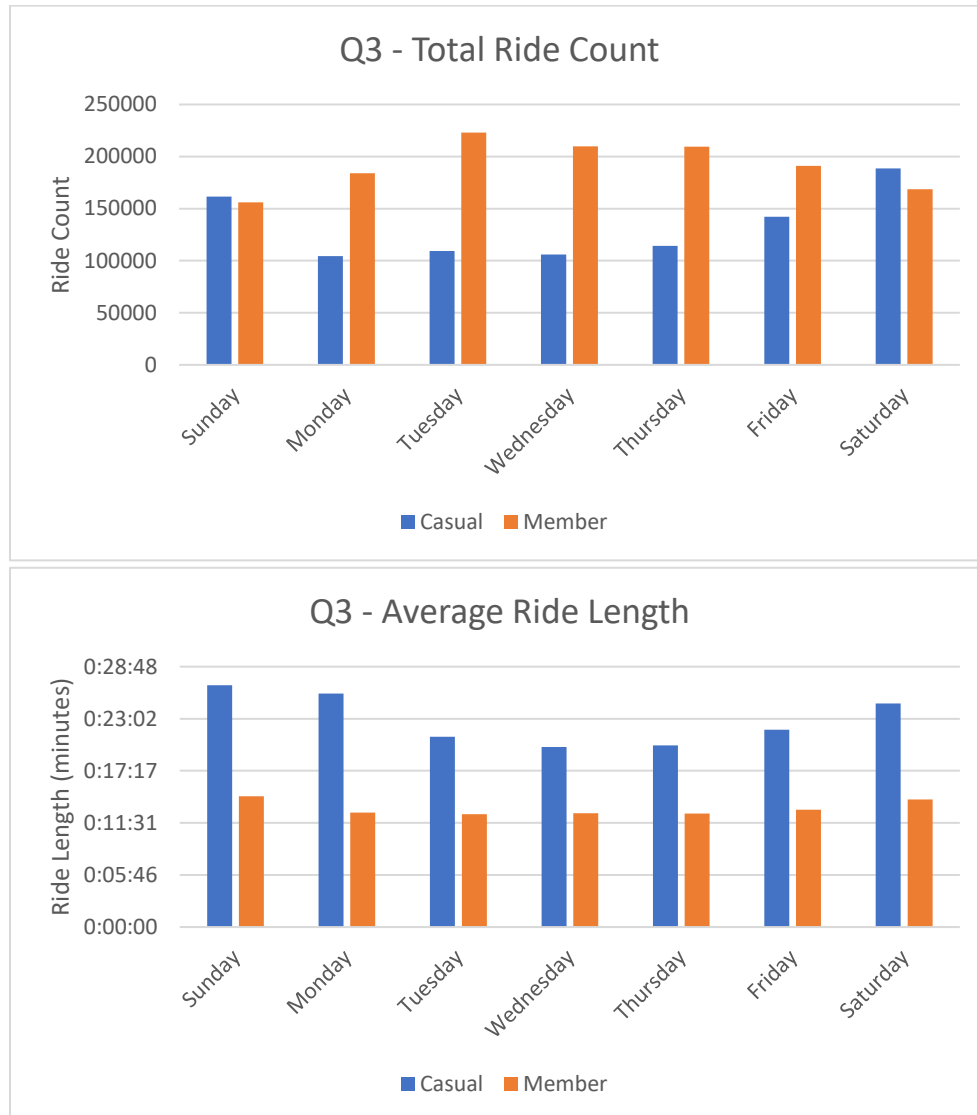
Q2



Ridership increases significantly in Q2 across both groups, marking the transition towards peak usage season. Member ride counts this quarter make up over 27% of yearly totals, with casual counts making up just over 29%. Casual riders show a shift toward weekend activity, with a noticeable increase in their average ride duration over members.

This marks the beginning of high engagement conditions, where increased casual usage creates a larger pool of potential membership conversions.

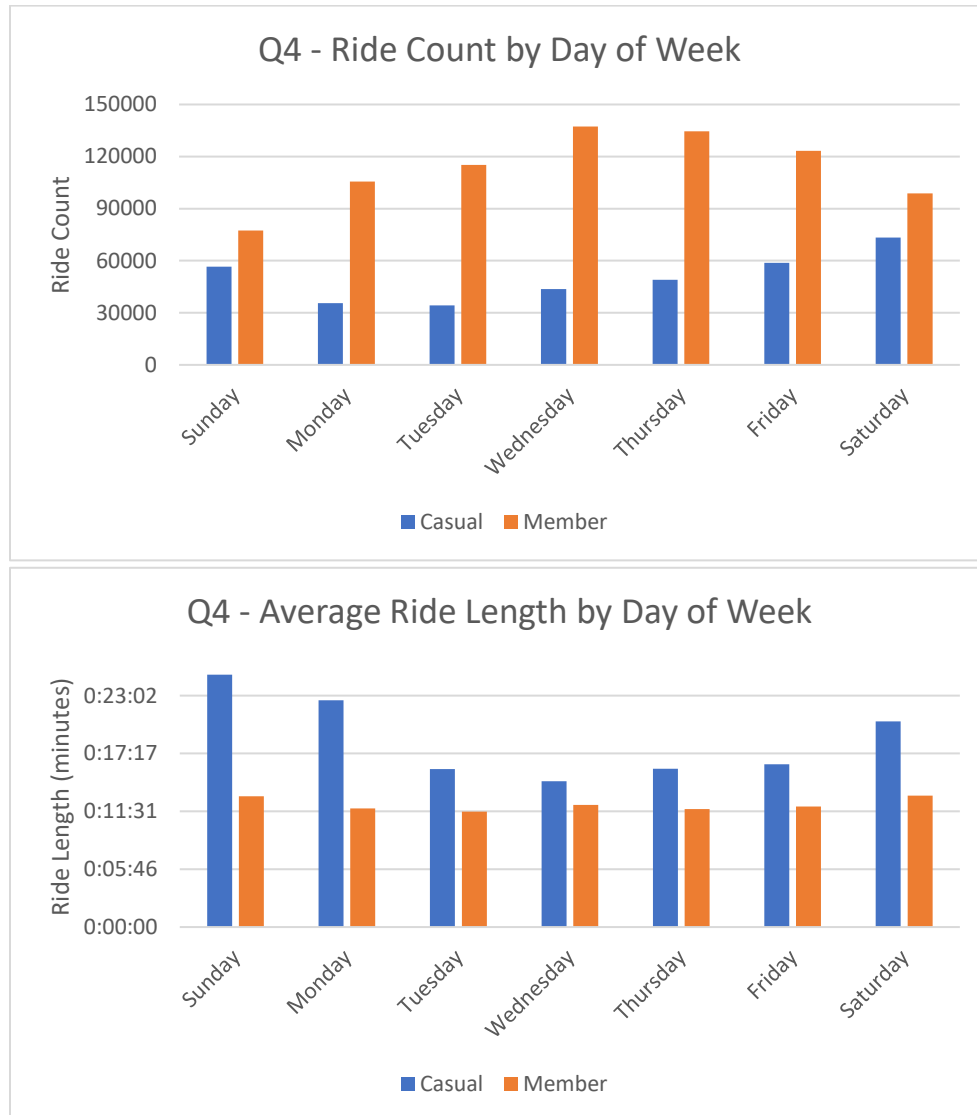
Q3



Q3 represents the peak of annual ridership, with the highest ride volumes observed for both members and casual rides. Member rides make up roughly 38% of the annual total, and casual rides over 46% of the annual total. While member behavior remains consistent, casual riders continue to concentrate on usage more on weekends, with longer ride durations.

The sustained high volume of casual riders during this period presents the strongest opportunity for conversion through targeted marketing efforts.

Q4



Ridership declines significantly in Q4, particularly among casual riders, while member usage decreases more moderately. Member rides this quarter make up roughly 22% of the annual total, and casual rides just under 18%. Despite the overall drop in volume, established behavioral patterns in riders persist, with member riders maintaining more routine usage and casual riders more weekend focused usage.

This decline in overall ride volume highlights the seasonal limitations of casual engagement and suggests reduced effectiveness of acquisition focused campaigns during this period.

Year End Summary

Column1	Casual Ride Length	Member Ride Length	Casual Ride Count	Member Ride Count
Sunday	0:25:19	0:13:05	333244	382346
Monday	0:22:52	0:11:33	228745	501006
Tuesday	0:18:37	0:11:34	225478	561053
Wednesday	0:17:18	0:11:31	221138	548527
Thursday	0:17:14	0:11:31	257141	573811
Friday	0:18:56	0:11:55	318032	526618
Saturday	0:22:01	0:12:56	413510	449151
Average/Totals	0:20:20	0:12:01	1997288	3542512

Key behavioral differences between casual and member riders persist throughout the year.

Casual riders primarily ride on weekends, especially Saturdays, indicating usage patterns aligned with leisure and recreational activities.

Member riders more frequently engage during weekdays, reflecting more routine, utility-based use of the service.

Casual riders consistently have longer average ride durations, while members take shorter, more frequent trips.

The higher ride frequency among members, combined with shorter durations, suggests fewer usage constraints and greater integration into daily travel patterns.

Recommendations and Insights

The analysis reveals clear behavioral differences between casual riders and members that can be leveraged to drive membership conversion:

- Target potential commuters: Members primarily use the service for consistent weekday travel, suggesting that casual riders with similar usage patterns are the most likely to convert. Marketing efforts should focus on positioning membership as a reliable and cost-effective commuting solution.
- Emphasize cost efficiency: Casual riders tend to take longer, less frequent rides, indicating sensitivity to per-ride costs. Campaigns should highlight the long-term savings and unlimited access associated with membership to reduce perceived cost barriers. Offering weekday discounts or promotions could likely lead to increased casual usage and conversion.
- Prioritize peak seasons: Ridership increases significantly during Q2 and Q3, when casual engagement is highest. Marketing campaigns deployed during these periods will reach a larger audience and are more likely to generate conversions.

Impact and Next Steps

The insights from this analysis provide a clear path to improving membership conversion by aligning marketing efforts with observed rider behavior. Targeting high potential casual riders, particularly those exhibiting frequent or weekday usage, can increase conversion rates while improving the efficiency of marketing costs.

As a next step, these findings should be interpreted into targeted marketing initiatives, such as campaigns focused on commuting cases and cost saving benefits during peak ridership periods. The time frames in which these campaigns are launched should also be carefully curated from this analysis to engage customers during the peak ridership periods of Q2 and Q3 to ensure cost effectiveness and efficiency.

From a data perspective, future analysis would benefit from the inclusion of rider-level identifiers and external factors such as weather and local events. These additional variables within data sets would allow for more precise segmentation and understanding of rider behaviors, supporting more effective and personalized conversion strategies.